

Collections Recovery Analysis

1: EXECUTIVE SUMMARY

COLLECTIONS RECOVERY ANALYSIS - EXECUTIVE SUMMARY

QUESTION ASKED:

Is the reported 11% month-on-month recovery improvement real?

ANSWER:

NO - The 11% claim is OVERSTATED. Real improvement is 0-3%.

KEY FINDINGS:

The 11% growth consists of:

- Data Quality Artifacts: 5-7% (duplicate payments, unvalidated PTPs)
- Portfolio Mix Effects: 2-3% (easier accounts added, harder written off)
- Targeting Changes: 2-3% (shifted to DPD>=30 from DPD>=60)
- Genuine Operational Improvement: 0-2% (actual improvement)

EVIDENCE DISCOVERED:

- 347 duplicate payment records (₹8M artificial recovery)
- 1,400 unvalidated promises-to-pay (metric inflation)
- 2,400 calls with multiple disposition records (attribution confusion)
- Cohort analysis shows same accounts flat/declining (Simpson's Paradox)
- Data quality score: 72/100 (acceptable with cleaning)

INVESTMENT RECOMMENDATION:

AI Voice Automation (₹10 Cr investment)

- Expected Annual Return: ₹2.5-3.2 Cr incremental
- Payback Period: 3.3 years
- 5-Year NPV: ₹4.2 Cr
- Confidence: 70%
- Why This: Highest ROI, largest addressable market, fastest payback

OVERALL CONFIDENCE LEVEL: 75%

(95% on duplicate payments, 70% on ROI projections)

2: KEY FINDINGS

DETAILED FINDINGS (7 Major Issues Found)

FINDING 1: DUPLICATE PAYMENTS

Status: HIGH PRIORITY

Records Affected: 347 duplicate payment records

Business Impact: ₹8M artificial recovery boost (1.3% of total)

Root Cause: Payment gateway retries or system reprocessing

Treatment: Deduplicated on payment_reference, kept first occurrence

Confidence: 95%

FINDING 2: UNVALIDATED PROMISES-TO-PAY

Status: HIGH PRIORITY

Records Affected: 1,400 out of 4,200 "kept" PTPs

Business Impact: PTP "kept" rate overstated by 23 percentage points

Root Cause: Agents marking PTP "kept" before payment verification

Treatment: Cross-validated PTPs with actual payments, 67% verification rate

Confidence: 70%

FINDING 3: MULTI-DISPOSITION CALLS

Status: HIGH PRIORITY

Records Affected: 2,400 calls (6.9%) with multiple disposition records

Business Impact: Attribution confusion, unclear which disposition is truth

Root Cause: Agent corrections, late-arriving data, or multi-agent handling

Treatment: Used FIRST_VALUE to keep original disposition

Confidence: 75%

FINDING 4: PORTFOLIO MIX EFFECTS (SIMPSON'S PARADOX)

Status: MEDIUM PRIORITY

Records Affected: Entire portfolio (30,001 accounts)

Business Impact: Aggregate metric improved 1% but cohorts flat/declining

Root Cause: Easier accounts added mid-year, harder accounts written off

Treatment: Cohort-based analysis controls for portfolio composition

Confidence: 80%

FINDING 5: TARGETING STRATEGY CHANGES

Status: MEDIUM PRIORITY

Records Affected: Daily targeting data (45,001 records)

Business Impact: 2-3% of recovery growth from easier targets

Root Cause: DPD threshold shifted from ≥ 60 to ≥ 30

Treatment: Controlled for targeting in root cause analysis

Confidence: 75%

FINDING 6: TIMEZONE INCONSISTENCIES

Status: MEDIUM PRIORITY

Records Affected: ~8,000 calls misclassified by timezone

Business Impact: Hour-level RPC calculation errors

Root Cause: 3 timezones (UTC, IST, Dubai) mixed in single field

Treatment: Normalized all to UTC, kept original timezone for reference

Confidence: 65%

FINDING 7: CAMPAIGN REDEFINITION

Status: MEDIUM PRIORITY

Records Affected: 3 campaigns with same name, different definitions

Business Impact: Time-series analysis unreliable, trends unmeasurable

Root Cause: Campaign definitions evolved, names reused

Treatment: Used campaign_id (not name) as truth, versioned changes

Confidence: 90%

3: DATA QUALITY ISSUES

DATA QUALITY ASSESSMENT REPORT

Overall Quality Score: 72/100 (Grade: C+)

Status: Acceptable with heavy cleaning required

Issue | Severity | Records | % of Total | Impact | Confidence

Duplicate Payments

HIGH | 347 | 1.4% | ₹8M artificial | 95%

Unvalidated PTPs

HIGH | 1,400 | 5.5% | Metric inflation | 70%

Multi-Dispositions

HIGH | 2,400 | 6.9% | Attribution broken | 75%

Timezone Errors

MEDIUM | 8,000 | 31% of calls | Hour-level errors | 65%

Agent ID Conflicts

MEDIUM | 50 agents | 0.2% | Tenure broken | 80%

Campaign Redefinition

MEDIUM | 3 campaigns | Variable | Trends unreliable | 90%

Late-Arriving Events

MEDIUM | 800 | 0.3% | Batch update issues | 85%

QUALITY BY COMPONENT:

Completeness (missing values): 92% - Excellent

Uniqueness (duplicates): 70% - Needs work

Consistency (format, range): 68% - Needs work

Accuracy (validated against truth): 75% - Acceptable

Timeliness (data freshness): 80% - Good

4: INVESTMENT RECOMMENDATION

WHERE TO INVEST ₹10 CRORE?

RECOMMENDATION: AI VOICE AUTOMATION

Investment Amount: ₹10 Cr

Expected Annual Return: ₹2.5-3.2 Cr incremental recovery

Payback Period: 3.3 years

5-Year NPV: ₹4.2 Cr @ 10% discount rate

Confidence Level: 70%

WHY THIS INVESTMENT:

1. HIGHEST ROI among alternatives

- 3.3-year payback vs 6-8+ years for other options
- ₹4.2 Cr positive NPV over 5 years

2. LARGEST ADDRESSABLE MARKET

- 1,000+ calls/day currently
- AI can scale to 3,000+ calls/day
- Digital channels can't match this volume

3. FASTEST IMPLEMENTATION

- 12 weeks to full deployment
- No hiring/training delays
- Immediate scale

4. NATURAL STRATEGIC FIT

- Voice is highest ROI channel
- Current RPC rate 35%, target 50%
- AI handles first-contact easily

ROI CALCULATION:

Current State:

- 1,000 calls/day × 35% RPC = 350 answered calls

- $350 \times 8\%$ conversion = 28 collections/day
- $28 \times ₹15,000$ avg recovery $\times 20$ days = ₹84 Lakhs/month
- Annual: ₹1,008 Lakhs

With AI Automation:

- 3,000 calls/day $\times 50\%$ RPC = 1,500 answered calls
- $1,500 \times 8\%$ conversion = 120 collections/day
- $120 \times ₹15,000$ avg recovery $\times 20$ days = ₹360 Lakhs/month
- Annual: ₹4,320 Lakhs

Incremental Return: ₹3,312 Lakhs/year (₹3.3 Cr/year)

DOWNSIDE SCENARIO:

- Borrower resistance (50% friction): -20% conversion
- Actual uplift only 1.5x (vs 4x assumed): = ₹1.5 Cr/year
- Still positive NPV, 6-year payback

WHY NOT OTHER OPTIONS?

Option	ROI	Payback	Why Not	Confidence
AI Voice	High	3.3y	← BEST	70%
Better Targeting	Medium	6.7y	Smaller market	60%
More Agents	Low	8.3y	Turnover issues	55%
Field Operations	Very Low	12.5y	Too expensive	50%
Telephony	Lowest	25y	Not bottleneck	40%

5: FILES & LINKS

ALL DELIVERABLE FILES

1. ARCHITECTURE_DESIGN.md
2. EXECUTIVE_MEMO.md
3. FORENSIC_ANALYSIS_DETAILED.md

ALL CODE FILES

1. Main_queries.sql - 350+ lines of production SQL
2. Python_code.py - Python analysis framework

6: ABOUT THIS ANALYSIS

PROJECT OVERVIEW

Title: Collections Recovery Analysis

Duration: ~20 hours of equivalent analytical work

Data Period: January 2026 - December 2026 (12 months)

Dataset Size: 639,346 records across 17 tables

ASSIGNMENT BRIEF:

Validate a business claim: "Recovery has improved by 11% month-on-month"

APPROACH:

Systematic data quality investigation → forensic analysis → root cause

attribution → counterfactual modeling → business recommendation with ROI

KEY STATISTICS:

Data Coverage:

- 17 datasets analyzed
- 639,346 total records
- 30,001 borrowers
- 30,001 agents
- 120+ campaigns
- Multi-channel (voice, SMS, WhatsApp, field)
- Multi-geography (India, UAE)

Issues Found:

- 7 major data quality findings
- 347 duplicate payment records
- 1,400 unvalidated PTPs
- 2,400 multi-disposition calls
- ₹8M+ in artificial inflation identified

Confidence Levels:

- Overall: 75%

- Duplicate payments: 95%
- Portfolio mix effects: 80%
- Real improvement estimate: 70%
- ROI projection: 70%

METHODOLOGY:

1. Data Quality Investigation

- └ Active search for problems
- └ Found 7 major issues
- └ Quantified business impact

2. Cohort Analysis

- └ Controlled for portfolio mix
- └ Showed Simpson's Paradox
- └ Revealed accounts declining cohort-by-cohort

3. Root Cause Attribution

- └ Data artifacts: 5-7%
- └ Portfolio mix: 2-3%
- └ Targeting changes: 2-3%
- └ Genuine improvement: 0-2%

4. Investment Analysis

- └ Ranked 5 alternatives by ROI
- └ Recommended AI voice automation
- └ Calculated 3.3-year payback

GITHUB LINK:- <https://github.com/utkarshvyas70/Collections-Analytics-Analysis>